# Varshap Walia

+1-514-569-8153 | varshwalia@gmail.com | varshapwalia.com | github.com/varshapwalia | linkedin.com/in/varshapwalia

**SUMMARY -** Software Engineer experienced in distributed systems, scalable backend platforms, cloud infrastructure, and production-grade LLM workflows with a strong focus on reliability, performance optimization, and AI-driven automation.

## Technical Skills

- **Languages:** Java, Python, C#/.NET, Go, JavaScript
- **Backend:** Kafka, Redis, REST APIs, gRPC, JWT, OAuth, OIDC, SAML, SCIM, Event-driven Architecture
- **Frameworks:** .NET / ASP.NET, Spring Boot, Django, FastAPI, Node.js, React
- **AI:** LLM Agents, RAG, LangGraph, LlamaIndex, FAISS, Vector Databases, Prompt Engineering, PyTorch
- **Cloud:** AWS, GCP, Kubernetes, Docker, Helm, Terraform, GitLab CI/CD, GitHub Actions, Kong Gateway
- **Databases & Observability:** PostgreSQL, MongoDB, MySQL, Elasticsearch, Datadog, New Relic, Sentry

## Experience

**Software Developer** — Jan. 2025 – Present
*Thales Group* — *Ottawa, ON*

- Developed and scaled distributed **IAM microservices** in **Java and .NET** supporting **SAML, OAuth, and OIDC** across multi-tenant environments on **Kubernetes and GCP**, serving **1M+ users**.
- Reduced p95 latency in critical **IAM workflows** from $\sim$**800ms** to under **120ms (**$\sim$**6**$\times$ **improvement)** through query optimization, indexing, and distributed **Redis** caching.
- Built autonomous **CI/CD failure recovery and test self-healing system** in **Python** using **LangGraph-based LLM agents** with tool execution and browser control, automatically resolving $\sim$**35–40%** of flaky test failures.
- Built **RAG-based incident analysis platform** integrating Jira, logs, traces, and metrics using **FAISS retrieval**, reducing MTTR by an estimated **20–30%.**
- Built distributed **load testing framework** using **Kubernetes, Helm, Terraform, and Locust** on **GCP**, simulating **100K+ concurrent users** and surfacing bottlenecks before production deployment.
- Led **LLM-based .NET 5** $\rightarrow$ **.NET 8 migration framework** for core feature services with multi-pass validation and feedback loops, achieving $\sim$**70%** to $\sim$**82%** automation.
- Optimized **LLM inference pipelines** using retrieval, context pruning, and multi-layer caching, reducing token consumption by up to **3–4**$\times$.

**SDE II** — Feb. 2022 – Aug. 2022
*Vahak* — *Bengaluru, IN*

- Redesigned geolocation platform built on **Go, Elasticsearch, and OpenStreetMap data**, expanding points-of-interest coverage by **4**$\times$ while improving data integrity and matching accuracy.
- Improved search relevance by $\sim$**89%** through relevance tuning and query analysis, increasing conversion.
- Built **Kafka-based pipelines** processing $\sim$**2M events/day** into MongoDB and S3 for analytics and ML workflows.

**Software Engineer** — Apr. 2019 – Jan. 2022
*Pinkblue* — *Bengaluru, IN*

- Identified manual handoff steps inflating delivery error rates; redesigned logistics workflow with **automated validation gates**, reducing errors $\sim$**17%** and saving an estimated **$187K annually**.
- Architected and built **POS and logistics platform** using **Python, Django, PostgreSQL, and AWS** across **100+ hospitals and 450+ pharmacies**, unifying inventory, order processing, and fulfillment as the **first system to span both verticals**.
- Led **GCP** $\rightarrow$ **AWS migration** leveraging **EC2, S3, EKS, MSK, SNS, Redshift, and OpenSearch**, reducing downtime incidents by $\sim$**23%** and improving observability with New Relic and Sentry.
- Built **real-time pricing engine** using market and vendor data feeds, improving order conversion by $\sim$**15%.**

**Backend Engineer** — Nov. 2017 – Mar. 2019
*Inclov Technologies* — *Gurgaon, IN*

- Built **encrypted messaging and push notification systems** on **Firebase** from scratch; improved delivery reliability from **72%** $\rightarrow$ **94%** and increased engagement by $\sim$**41%** over six months post-launch.
- Implemented **ML-based content moderation pipeline**, reducing manual moderation by $\sim$**27%.**

## Education

**Concordia University** — Montreal, QC
*Master of Applied Computer Science,* ***Major in AI*** — *Sept. 2022 – Dec. 2024*